

4.3 Evaluation Metrics:

We employ five widely-used metrics in finance to compare the investment rewards of FINMEM against other algorithmic trading agents. Here are their introductions:

- **Cumulative Return** [21]: Cumulative Return is a key trading performance metric because it provides a comprehensive insight into investment performance, especially for strategies that emphasize long-term growth and reinvestment. The effectiveness of different investment strategies is evaluated based on their Cumulative Returns, which reflect the total change in value over time. In this study, we compute Cumulative Returns over the specified period by summing daily logarithmic returns, as outlined in Equation 7. This method is widely accepted in the finance area due to its ability to precisely capture minor price fluctuations and symmetrically address gains and losses. In essence, a higher Cumulative Return typically indicates a more effective strategy.

$$\begin{aligned}\text{Cumulative Return} &= \sum_{t=1}^n r_i \\ &= \sum_{t=1}^n \left[\ln \left(\frac{p_{t+1}}{p_t} \right) \cdot \text{action}_t \right],\end{aligned}\quad (7)$$

where r_i represents the logarithmic return for day $t + 1$, p_t is the closing price on day t , p_{t+1} is the closing price on day $t + 1$, and action_t denotes the trading decision made by the model for that day.

- **Sharpe Ratio** [43]: Sharpe Ratio is another core metric for evaluating investment performance and adjusting returns for risk. It is calculated by dividing the portfolio’s average excess return (R_p) over the risk-free rate (R_f) by its volatility (σ_p), as shown in Equation 8. This metric adjusts returns for risk, with a higher ratio indicating better risk-adjusted performance. Essential in comparing different portfolios or strategies, it contextualizes performance against similar investments. Although a Sharpe Ratio above 1 is typically considered favorable and above 2 as excellent, these benchmarks can vary depending on the context of comparison.

$$\text{Sharpe Ratio} = \frac{R_p - R_f}{\sigma_p} \quad (8)$$

- **Annualized Volatility and Daily Volatility**[8]: Annualized Volatility (Annum-Volatility) is calculated as the **Daily Volatility** (standard deviation of daily logarithmic returns) multiplied by the square root of the typical number of trading days in a year (252) as outlined in Equation 9, is vital for assessing investment risk. This measure reflects the extent of fluctuation in a security or market index’s returns over a year, indicating potential deviations from average returns. It’s especially relevant for investors with specific risk profiles, such as those who are risk-averse, who may prefer portfolios demonstrating lower annualized volatility.

$$\text{Annum-Volatility} = \text{Daily Volatility} \times \sqrt{252} \quad (9)$$

- **Max Drawdown** [1]: Max Drawdown is a metric for assessing risk. It represents the most significant decrease in a portfolio’s value, from its highest (P_{peak}) to its lowest point (P_{trough}) until a new peak emerges, detailed in Equation 10. Indicative of investment strategy robustness, a smaller Max Drawdown suggests reduced risk.

$$\text{Max Drawdown} = \max \left(\frac{P_{\text{peak}} - P_{\text{trough}}}{P_{\text{peak}}} \right) \quad (10)$$

In our experiments and ablation studies, we recorded the metric outcomes as an average from five repeated trials.

5 Experiments:

5.1 Implementation Details:

In the Trading Agents Comparison, FINMEM employs GPT-4-Turbo as its backbone algorithm. The temperature parameter of the model was set at 0.7 to maintain a balance between response content consistency and model creativity.

³The bold numbers in this and subsequent tables signify the best performance for the respective metrics.

Ticker	Model	Cumulative Return (%)	Sharpe Ratio	Daily Volatility (%)	Annualized Volatility (%)	Max Drawdown (%)
TSLA	Buy and Hold	-18.6312	-0.5410	4.4084	69.9818	55.3208
	FINMEM	61.7758*	2.6789	2.9522	46.8649	10.7996
	Generative Agents	13.4636	0.5990	2.8774	45.6774	24.3177
	FinGPT	-7.4554	-0.2795	3.4145	54.2027	42.3993
	A2C	13.7067	0.3979	4.4096	70.0009	52.3308
	PPO	1.2877	0.0374	4.4110	70.0232	54.3264
	DQN	33.3393	0.9694	4.4027	69.8900	52.0033
NFLX	Buy and Hold	35.5111	1.4109	3.1964	50.7410	20.9263
	FINMEM	36.4485*	2.0168	2.2951	36.4342	15.8495
	Generative Agents	32.0058	1.5965	2.5460	40.4168	16.9893
	FinGPT	9.0090	0.4266	2.6819	42.5732	28.2705
	A2C	14.6155	0.5788	3.2071	50.9112	25.0184
	PPO	8.4121	0.3330	3.2086	50.9344	25.0184
	DQN	-12.2067	-0.4833	3.2078	50.9217	28.7017
AMZN	Buy and Hold	-10.7739	-0.4980	2.7697	43.9674	33.6828
	FINMEM	4.8850*	0.2327	2.6872	42.6576	22.9294
	Generative Agents	-13.9271	-0.9981	1.7864	28.3576	27.7334
	FinGPT	-29.6781	-2.1756	1.7464	27.7225	28.4838
	A2C	-6.3591	-0.2938	2.7706	43.9819	26.1275
	PPO	-8.4194	-0.3891	2.7702	43.9761	33.6828
	DQN	-29.9820	-1.3906	2.7603	43.8177	38.3740
MSFT	Buy and Hold	14.6949	0.8359	2.2326	35.4411	15.0097
	FINMEM	23.2613*	1.4402	2.0512	32.5617	14.9889
	Generative Agents	-18.1031	-1.6057	1.4318	22.7285	24.2074
	FinGPT	5.7356	0.4430	1.6442	26.1008	12.8459
	A2C	0.4598	0.0261	2.2357	35.4913	23.6781
	PPO	12.8067	0.7282	2.2333	35.4532	19.5355
	DQN	14.7397	0.8385	2.2326	35.4408	25.1845
COIN	Buy and Hold	-30.0071	-0.5150	6.7517	107.1795	60.5084
	FINMEM	34.9832*	0.7170	5.6538	89.7515	35.7526
	Generative Agents	3.4627	0.0896	4.4783	71.0908	32.0957
	FinGPT	-88.7805	-1.9507	5.2736	83.7153	73.5774
	A2C	-	-	-	-	-
	PPO	-	-	-	-	-
	DQN	-	-	-	-	-

Table 2: Overall trading performance comparison during testing period between FINMEM and other algorithmic agents across five stocks.* indicates that the result of the Wilcoxon signed-rank test is statistically significant.³

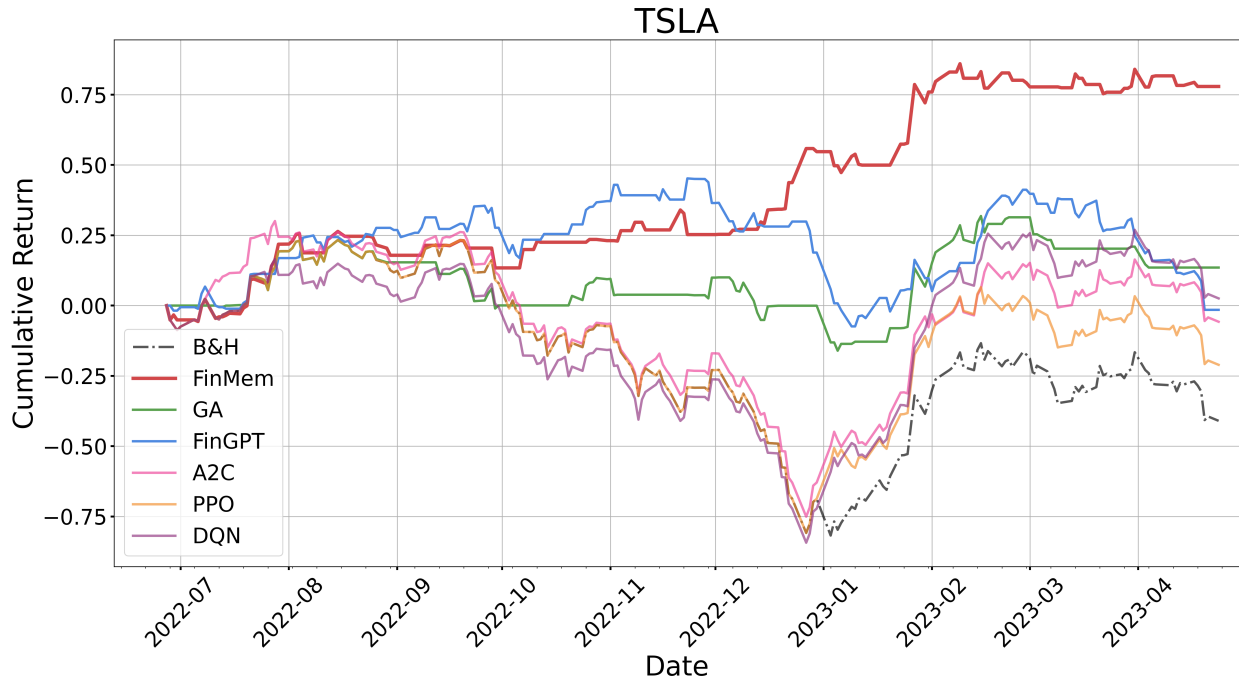


Figure 7: Cumulative Return of FINMEM on trading Tesla, Inc. (TSLA) stock Over an Extended Testing Period.

It was trained on financial data from August 17, 2021, to October 05, 2022, and underwent testing with data from October 06, 2022, to April 10, 2023. The training period was chosen to account for the seasonal nature of corporate financial reporting and the duration of data retention in FINMEM’s memory module. The selected training duration ensures the inclusion of at least one publication cycle of either Form 10-Q, classified as intermediate memory, or Form 10-K, regarded as deep memory, or in some instances, both. This strategy ensures that the experiences retained in FINMEM are still influential during the testing phase for a significant period. Additionally, the training duration allowed FINMEM sufficient time to establish inferential links between financial news, market indicators, and stock market trends, thereby accumulating substantial experience. Furthermore, we set the number of top memory events retrieved from each layer of long-term memory at 5. We ran FINMEM using each of the three available risk inclination settings. The reported performance outcomes are based on the setting that achieved the highest cumulative return during the testing phase.

To maintain consistency in the comparison, the training and testing phases for the other two LLM-based agents were aligned with those of FINMEM. For parameters of other LLM-based agents that are not encompassed by FINMEM’s configuration, they were kept in accordance with their original settings as specified in their respective source codes.

Considering that DRL algorithms need extensive training data for stable and converged results, and given our daily evaluation of trading performance, we extended the DRL agents’ training period to roughly a 10-year span, from January 1, 2012, to October 05, 2022, for a fair comparison. The testing period was kept consistent with the other models. The DRL algorithms were implemented using Stable Baselines 3 [41].

FINMEM’s performance was benchmarked against that of the most effective comparative model, using Cumulative Return and Sharpe Ratio as the primary evaluation metrics. The statistical significance of FINMEM’s superior performance was ascertained through the non-parametric Wilcoxon signed-rank test, which is particularly apt for the non-Gaussian distributed data.

5.2 Algorithmic Trading Agents Comparison (RQ1 & RQ2)

In this experiment, we assess the stock trading performance of FINMEM against other models, focusing on stocks from five companies in different trading sectors: Tesla, Inc. (TSLA), Netflix, Inc. (NFLX), Amazon.com, Inc. (AMZN), Microsoft Corporation (MSFT), and Coinbase Global, Inc. (COIN). The performance of all algorithmic trading agents across five key metrics is consolidated in Table 2. Given the pivotal role of Cumulative Return in evaluating trading performance over time, we present detailed time series plots in Figure 5 and Figure 6. It’s important to note that the trading performance of FINMEM for COIN was exclusively compared with LLM trading agents and the baseline. This is because Coinbase Global, Inc. completed its IPO in April 2021 and, as a result, had not accumulated enough trading data to facilitate stable outcomes with Deep Reinforcement Learning (DRL) algorithms. These plots illustrate the changes in Cumulative Return for each of the five companies throughout the testing phase, offering an in-depth comparison of performance.

In response to **RQ1**, the trading outcomes presented in Table 2 reveal that FINMEM outperforms all other algorithmic trading agents and the B&H baseline strategy in terms of Cumulative Return and Sharpe Ratio. FINMEM’s superiority is statistically significant when compared to the second-best trading strategy. Specifically, for TSLA and NFLX, FINMEM’s strategy achieves Sharpe Ratios exceeding 2.0 and Cumulative Returns surpassing 0.35 while maintaining the lowest Volatility and Max Drawdown. These indicators underscore FINMEM’s ability to generate higher returns per unit of risk. In the case of MSFT and NFLX, FINMEM also records a Sharpe Ratio above 1.0 and a Cumulative Return over 0.2, coupled with relatively low Volatility and Max Drawdown, demonstrating its impressive trading performance. For AMZN and COIN, FINMEM consistently delivers positive Cumulative Returns and superior Sharpe Ratios, outperforming other strategies that yield negative values for these metrics. Additionally, its Volatility and Max Drawdown are on the lower end. Hence, these results collectively demonstrate FINMEM’s robust trading performance across a diverse range of trading sectors. Specifically, FINMEM exhibits superior performance compared to the two other LLM agents in our study, FINGPT and the general-purpose generative agent developed by Park et al. This underscores the effectiveness of FINMEM’s unique profiling and memory structure, which are particularly tailored for LLM agents dealing with financial data, significantly enhancing their investment decision-making capabilities.

In response to **RQ2**, the main challenge for DRL trading agents is that they require training data with a large volume and extended time span, which are hard to achieve when operating on stocks with limited historical data. As shown in Table 2, our experiments reveal that FINMEM achieves superior trading performance with a much shorter training duration compared to DRL trading agents trained on data spanning nearly a decade. This efficiency makes FINMEM particularly useful for newly public companies like Coinbase Global, Inc., which have limited trading histories. DRL agents often face convergence issues due to inadequate training data in such cases. Moreover, even among LLM-based trading agents suited for shorter training periods, FINMEM’s performance stands out, as illustrated in Figure 6.